



Indian Association for the Cultivation of Science
(Deemed to be University under the *de novo* category)

Integrated Bachelor's-Master's Program

End Semester Examination - Autumn 2025

Subject: Introductory Mathematical Methods and
Classical Mechanics

Subject Code(s): PHS 1101
Time allotted: 3 hrs

Full marks: 50

Answer all questions

1 (a) Show that the motion of a particle of mass m in a central inverse square force $\vec{F} = \frac{c}{r^2}$ is a conic given by $\frac{1}{r} = -k/h^2 + A \cos(\theta - \theta_0)$, where h is the angular momentum per unit mass and $k = c/m$. Find the condition for this orbit to be an ellipse. (7 marks)

(b) Show that $\vec{\nabla} \times (\vec{\nabla} \phi) = 0$, where ϕ is a scalar function. (3 marks)

2 (a) Show that for a system of particles, the location of the centre of mass is independent of the choice of origin. (4 marks)

(b) Consider a system of particles of masses $m_1, m_2, \dots, m_n$ with total mass M . Show that the total angular momentum of the system about the origin is equal to the sum of angular momentum of a mass M (kept at the centre of mass) about the origin and the total angular momentum of all the particles about the centre of mass. Also show that the total angular momentum is conserved in absence of any external torque when the inter-particle force between any two particles acts along the line joining them. (6 marks)

3 (a) Consider a frame S' rotating with an angular velocity $\vec{\omega}$ with respect to an inertial frame S where the origins of the two frames coincide. Show that the acceleration vector between the two frames are related as,

$$\frac{d^2 \vec{r}}{dt^2} = \frac{d'^2 \vec{r}}{dt'^2} + \vec{\omega} \times (\vec{\omega} \times \vec{r}) + 2\vec{\omega} \times \frac{d' \vec{r}}{dt'} + \frac{d' \vec{\omega}}{dt'} \times \vec{r}$$

(symbols have their usual meaning) (7 marks)

(b) Find the moment of inertia of a rectangular sheet of sides a and b about an axis which coincides with the side of length a (Assume σ to be the mass per unit area). (3 marks)

4 (a) Consider a fluid in static equilibrium where the external body force per unit volume is $\vec{F}$. If $P(x, y, z)$ is the pressure at any point inside the fluid then show that the condition of equilibrium for the fluid is $\vec{F} = -\vec{\nabla} P$. Prove Pascal's Law for fluids from this equation. (5 marks)

(b) Define a streamline in the field of steady flow of an ideal fluid. Prove that at every point in a streamline,

$$\frac{v^2}{2} + \int \frac{dp}{\rho} + gh = \text{constant}$$

(symbols have their usual meaning).

(5 marks)

5 (a) Find the Fourier decomposition of a saw-tooth wave which rises from 0 to a over a time period T as $y(t) = at/T$ and then sharply falls to zero and then is repeated .

(6 marks)

(b) A point particle of mass m rolls down from the peak of a smooth spherical surface of radius R . If the potential energy is taken to be zero at the peak and the particle starts from rest, find the kinetic energy and the change in potential energy when the particle has rolled down by an angle θ with respect to the vertical passing through the peak and the origin of the sphere. Also find the angle at which the mass m leaves the spherical surface.

(4 marks)